

# Epidemic Data Analysis Based on Data Visualization Technology

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**Abstract**—The outbreak of COVID-19 at the end of 2019 has had a heavy impact on countries around the world, and its rapid spread has posed a huge challenge to the prevention and control work of all countries. Through data visualization technology, this paper comprehensively analyzes the epidemic data in China from multiple levels, including comparison analysis based on time span and comparison analysis based on countries. At the same time, this paper also analyzes the significant impact of the epidemic on China's national economy from the macro and micro levels. Finally, the linear regression model is used to predict the GDP data during the epidemic period, and the comparison between the real and predicted GDP data during the epidemic period is made. It was found that the epidemic had a great impact on China's national economy in the early stage, but the impact gradually weakened as time went by. Epidemic data analysis based on data visualization technology can provide powerful data support for epidemic prevention and control.

**Keywords**—data visualization; the linear regression; COVID-19; epidemic prevention and control; national economy

## I. INTRODUCTION

The global outbreak of COVID-19 since December 2019 has posed a severe challenge to global public health security. By May 2022, more than 500 million cases had been confirmed and more than 6 million people had died globally. The high transmission rate of COVID-19 and a series of problems it may cause make the medical needs of COVID-19 far higher than those of other diseases. Rising demand for medical care has led to a series of problems, such as poor access to medical services and low efficiency of medical services, which will make the epidemic prevention efforts even more difficult to some extent. In order to improve the work efficiency of epidemic prevention and control, and to provide effective medical service to patients, improve the epidemic data analysis efficiency is the key step. And visualization analysis of epidemic data can not only provide clear data support for the epidemic prevention and control work, but also can improve medical efficiency. The improvement of medical efficiency will help improve the cure rate of patients.

The analysis of any problem cannot be separated from the support of data. As for data visualization [1], people's first concept is all kinds of charts. In fact, chart is only the carrier of data visualization. Data visualization is still inseparable from the support of data. It carries data in the form of charts and presents data in the form of charts. Studies have shown that human eyes can recognize visual

symbols several orders of magnitude faster than digital and text information display methods. And, in everyday life, the main way people get data is through the visual system. Data visualization technology is to visually display the abstract information implied by numbers, texts and other information. Data visualization technology can help people understand and process all kinds of information more quickly.

Based on this, through data visualization technology, epidemic data are analyzed based on time span and comparison analysis based on countries. And, the significant impact of the epidemic on China's national economy from the macro and micro levels is analyzed too. Finally, this paper predicts and analyzes the GDP data during the epidemic period through the linear regression model. This paper focuses on how to analyze epidemic data through data visualization technology to provide support for epidemic prevention and control.

## II. RELEVANT THEORETICAL ANALYSIS

### A. Research on Epidemic Prevention and Control

The COVID-19 outbreak in late 2019 is a major public health emergency and has been assessed by the World Health Organization as the most dangerous infectious disease. Since the outbreak of COVID-19, epidemic prevention and control has been one of China's priorities. Scholars have conducted extensive studies on all aspects of epidemic prevention and control.

Lin Taoyu et al. (2020) on issues related to hospital nursing human resources under the state of epidemic prevention and control, the nursing emergency organization system, information sharing mode and scientific allocation of nursing human resources were studied to improve the nursing work efficiency [2].

Zhao Zhen et al. (2020) established a mode of "three districts and two consultations" to optimize the patient treatment process and effectively guarantee the effect of epidemic prevention and control in view of the significant increase of patients in the emergency department during the epidemic [3].

Zhang Tianchen et al. (2021) took epidemic prevention and control in Jiangxi Province as an example to construct a multi-stage SEIR model and evaluate the effectiveness of epidemic prevention and control in Jiangxi Province [4].

### B. Research on Data Visualization Technology

The development of data visualization technology can be traced back to the computer graphics in the 1950s. In that era, computers were used to make the first charts. With the

continuous development of data visualization technology, data visualization technology has been widely used in many fields, including e-commerce, meteorology, education and many other fields. Scholars at home and abroad have conducted in-depth research on data visualization technology and its application.

Qin Rufu et al. (2009) visualized ocean data based on four dimensions, including ocean surface and interior, seabed surface and deep, which helped to organize, manage and display multidimensional ocean data in a unified manner [5].

Liu Xiangdong et al. (2017) visualized the operation of related data of tourism e-commerce by using thermal maps through data visualization technology, and put forward targeted suggestions for e-commerce product promotion strategies based on data visualization results [6].

Liu Hai et al. (2018) realized the adjustment and improvement of classroom teaching, student learning and teaching management based on data visualization technology from the three levels of teachers, students and teaching managers [7].

### III. VISUALIZATION ANALYSIS OF EPIDEMIC DATA BASED ON PYTHON TECHNOLOGY

#### A. Data Visualization Techniques Based on Python

With the development and application of data visualization technology, there are many data visualization tools. Data visualization tools today cover a wide range of development tools for enterprises, developers, and general users. This also presents a difficult choice of data visualization tools. Python's open source features and rich library features combine the advantages of multiple data visualization tools. Using Python, we can easily use a variety of data visualization tools, which to some extent solves the problem of data visualization tool selection.

#### B. Process of Visualization Analysis of Epidemic Data

In essence, data visualization technology is to show the abstract information implied by numbers, texts and other information in a visual way. In fact, data visualization analysis includes a complete set of processes from data acquisition, data preprocessing, visual mapping and human-computer interaction. As the key to epidemic prevention and control, the data visualization analysis process needs to be more rigorous. The specific process of visualization analysis of epidemic data is shown in Figure 1.

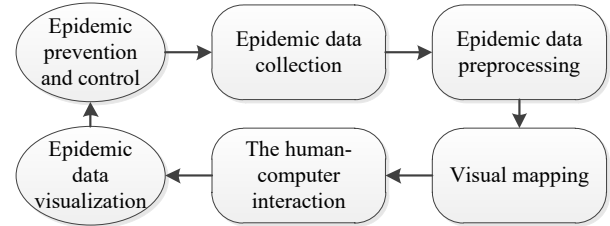


Figure 1. Visualization analysis of epidemic data process

#### C. Visualization Analysis of Epidemic Data in China

China's first COVID-19 case was reported in December 2019. It can be found through the visualization results of epidemic data in China in Figure 2. The epidemic in China developed rapidly in the early stage, and then stabilized, mainly due to the implementation of a series of prevention and control policies in China. However, since March 2022, the cumulative number of confirmed cases has increased significantly, mainly because a large number of confirmed cases have appeared in some regions of China since March 2022.

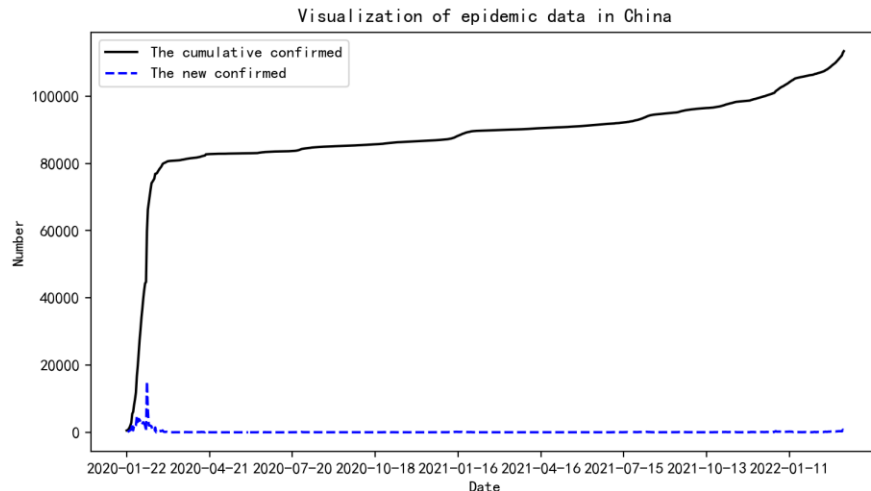


Figure 2. Visualization of epidemic data in China

#### D. Visual Comparative Analysis of Epidemic Data between China and the US

COVID-19 is not only breaking out in our country, but also in other countries. In addition, due to the differences in

epidemic prevention and control among countries, the epidemic prevention and control effects of different countries vary greatly. Taking China and the United States as an example, the visual comparative analysis of epidemic data between China and the United States is shown in Figure 3. In

the first three months after the outbreak of COVID-19, China's epidemic situation was relatively severe, but with the concerted efforts of the whole country, China's epidemic

prevention and control has achieved remarkable results. In the United States, the cumulative number of confirmed and deaths cases is increasing rapidly.

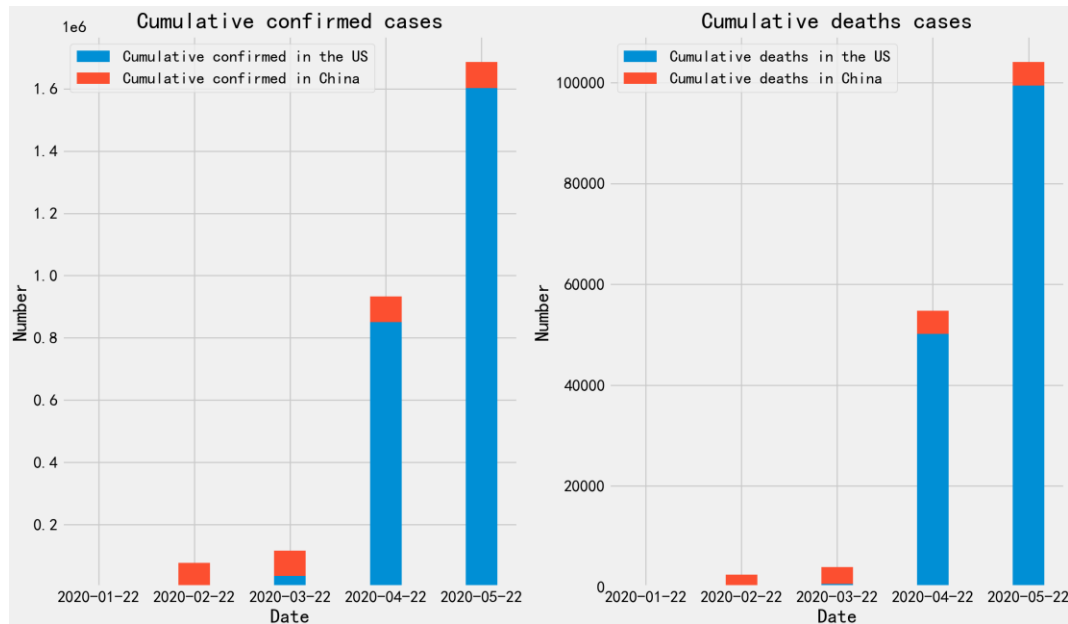


Figure 3. Visual comparison of epidemic data between China and the US

#### E. Visual Analysis of the Impact of the Epidemic on China's National Economy

Although China has prevented the large-scale spread of the epidemic through effective epidemic prevention and control measures, in the context of the severe global situation of COVID-19, how much is China affected by the epidemic?

The following is a visual analysis of the relevant data of China's national economy during the epidemic.

Taking China's GDP data and China's industrial export price index data from the year before the outbreak (2019) to the end of 2021 as examples, the impact of the epidemic on China's national economy is analyzed from a macro perspective.

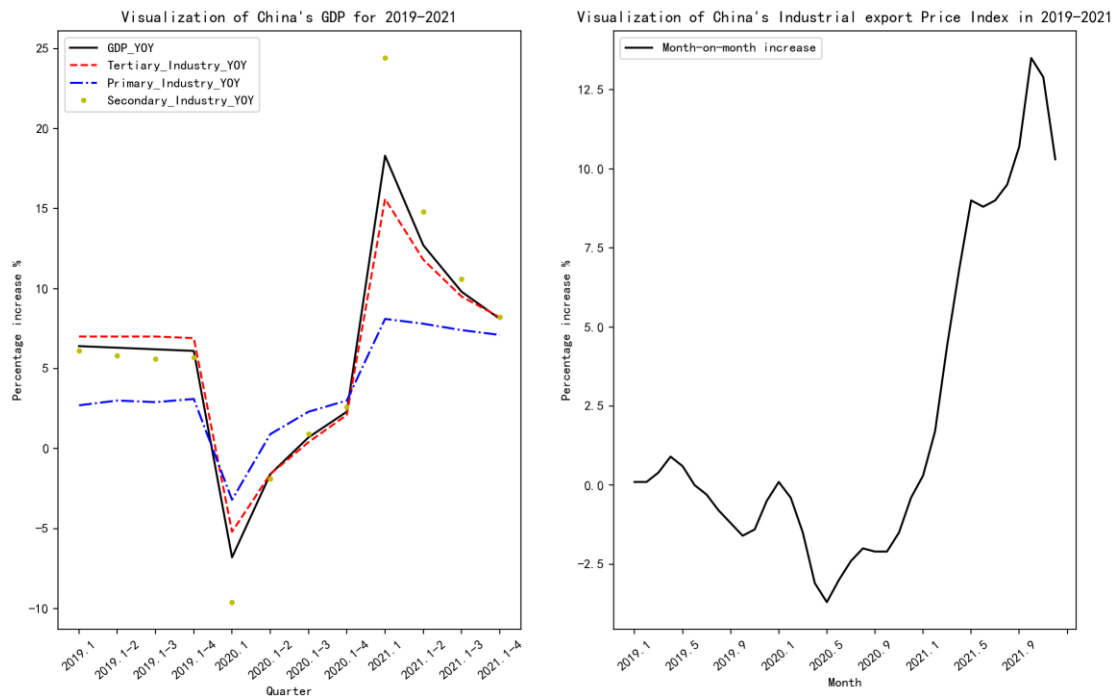


Figure 4. Visualization of the macro impact of the epidemic on China's national economy

YOY in the figure represents year-on-year growth. As can be seen from Figure 4, in the early stage of COVID-19 outbreak, both China's GDP and China's industrial export price index declined significantly, indicating that the epidemic had a huge impact on China's national economy at the macro level. The steady recovery after the initial outbreak indicates that China's epidemic prevention and control policies have achieved remarkable results.

The COVID-19 has had a profound impact not only on individual countries, but also on individuals. Consumer price index (CPI) is usually an important indicator to observe the level of inflation, which is related to all aspects of people's life. Taking China's consumer price index as an example, this paper analyzes the impact of the epidemic on China's urban and rural areas from the micro level.

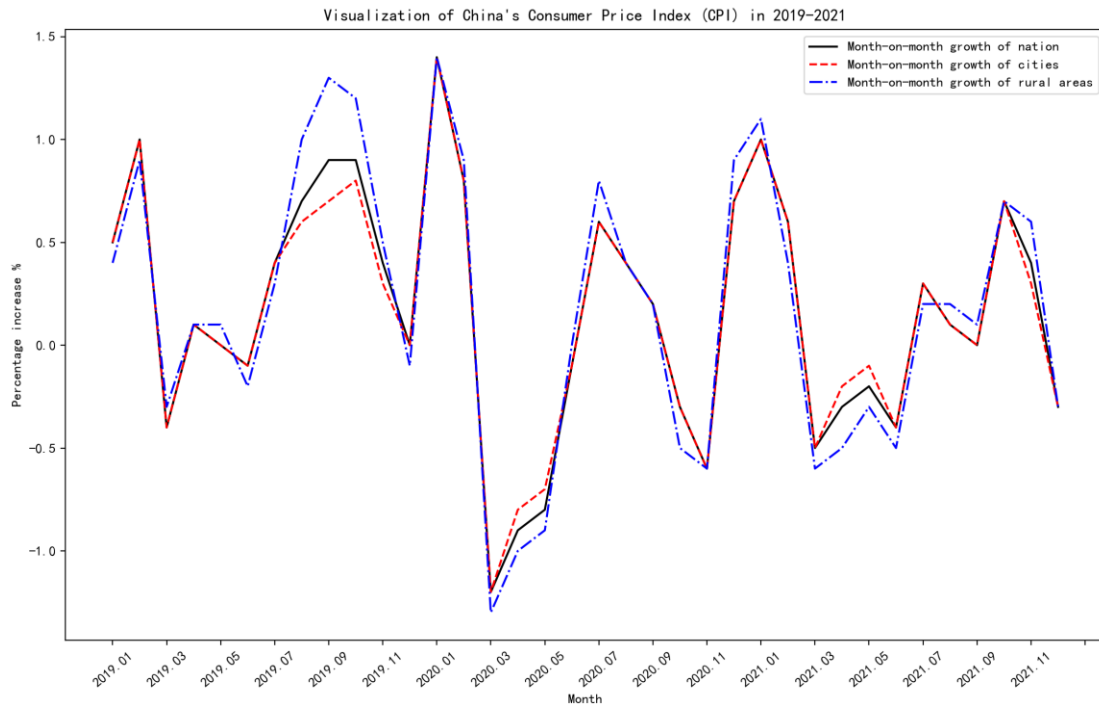


Figure 5. Visualization of the micro-impact of the epidemic on China's national economy

As can be seen from Figure 5, in the early stage of COVID-19 outbreak, the fluctuation of the consumer price index (CPI) in terms of national, urban and rural month-on-month growth was more obvious than in other periods. It can be seen that at the micro level, the epidemic has also had a huge impact on China's national economy. After the initial stage of the outbreak, the epidemic gradually recovered to the fluctuation range before the outbreak, which again shows that China's epidemic prevention and control policy has achieved remarkable results.

#### IV. GDP PREDICTION AND COMPARATIVE ANALYSIS BASED ON HISTORICAL DATA

Through the visual analysis of the macro and micro impacts of the epidemic on China's national economy, it can be found that the epidemic has a huge impact on China's national economy. So how much impact has the epidemic had on China's national economy? And whether the impact of the epidemic on China's national economy has weakened or intensified over time? These problems deserve further in-depth analysis. In this paper, the prediction method of machine learning technology will be used to predict the GDP data during the epidemic based on the historical GDP data before the epidemic. By comparing and analyzing the

predicted GDP data during the epidemic with the real GDP data during the epidemic, the impact of the epidemic on China's national economy and the changes in the impact of the epidemic on China's national economy over time will be explored.

This paper uses a linear regression model to predict the GDP data during the epidemic. It can be seen from the comparative analysis of real and predicted GDP during the epidemic of figure 6, the predicted GDP during the epidemic was relatively stable, indicating that the historical GDP data before the epidemic was also relatively stable. The real value of GDP fluctuated greatly during the epidemic, indicating that the epidemic had a significant impact on China's GDP.

In addition, it can be seen from the analysis of the difference between the real and predicted GDP during the epidemic of Figure 6, in the first year of the epidemic, there was a large error between the real GDP value and the predicted GDP value, with an overall error between -20% and -10%. In the second year of the epidemic, the error between the real and predicted GDP values gradually narrowed, and the overall error value was between -5% and 5%. This further shows that the epidemic prevention and control has achieved results, and China's national economy is gradually on the right track.

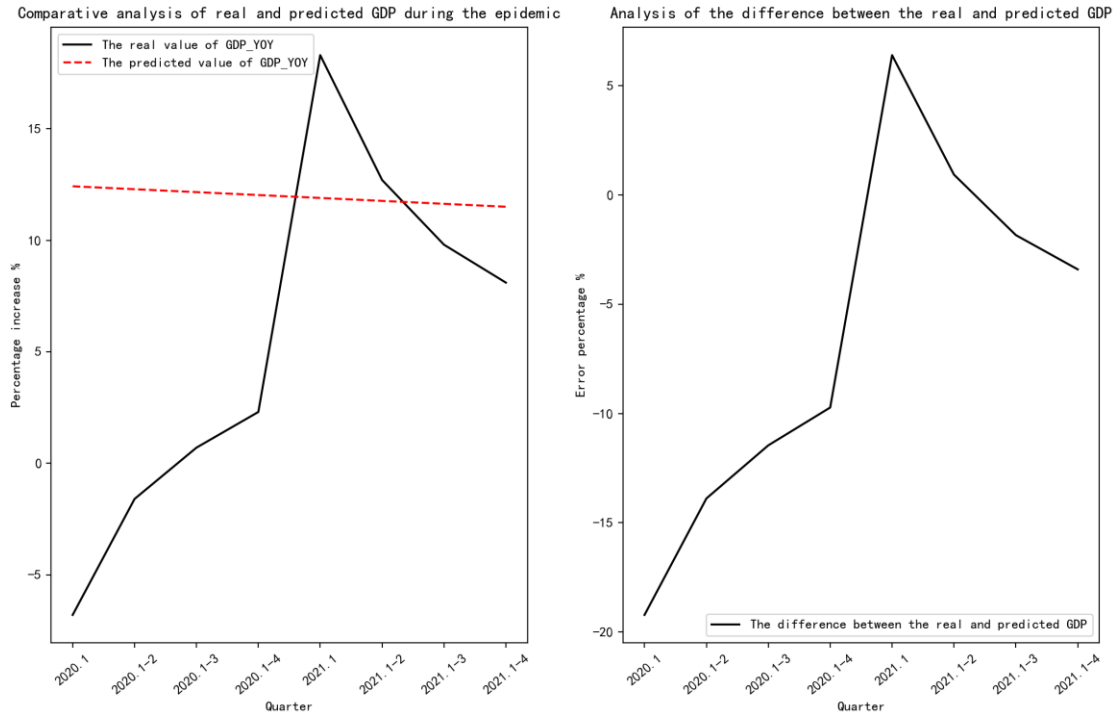


Figure 6. GDP prediction and comparative analysis based on historical data

## V. CONCLUSION

Based on data visualization technology, this paper provides an in-depth analysis of COVID-19 data from all levels. To find the supporting role of data visualization technology in epidemic prevention and control. By comparing the visualized results of China in different periods of epidemic outbreak and comparing the trend of epidemic data between China and the US through data visualization technology, the effectiveness of China's epidemic prevention and control work was analyzed. At the same time, the impact of COVID-19 on China's national economy was analyzed from both macro and micro levels through data visualization technology. Finally, the linear regression model was used to predict the GDP data during the epidemic, and a comparative analysis was made between the real and predicted GDP values during the epidemic. It was found that the epidemic had a great impact on China's national economy in the early stage, but the impact gradually weakened as time went by.

Although this paper analyzes the epidemic data from multiple levels through data visualization technology, it lacks further mining of the hidden information behind the data, which is also the direction of future research.

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